Department of CSE(AIML), PES University
UE26CS151A -Python for Computational Problem Solving
Laboratory Week 6

Note: Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. The observation record must be signed by the faculty in charge. For every laboratory session, the observation component carries **marks**.

Problem Statements

Q1. Count the Number of Words in a Text File Write a program to read a text file and count the total number of words present in it.

- **User Input:** File name entered by the user.
- **Data Types:** Use string for text, int for word count.

Q2. Display Student Marks Using a Dictionary Write a program to store student names and their marks in a dictionary and display the name and marks of each student.

- **User Input:** Student names (string) and marks (int) entered by the user.
- **Data Types:** Use dict with string keys and int values.

Q3. Find the Highest Mark from a CSV File Write a program to read student names and their marks from **week_6_set1_q3.csv** and display the name of the student who scored the highest mark.

- **User Input:** No user input required.
- **Data Types:** Use string for names, int for marks.
- **File Reference:** Refer to **week_6_set1_q3.csv** for input.

Q4. Count the Frequency of Each Word in a Text File Write a program to read words from a text file and use a dictionary to count how many times each word appears.

- **User Input:** File name entered by the user.
- **Data Types:** Use dict with string keys and int values.

Q5. Display Products with Low Stock from a CSV File Write a program to read product names and their stock quantities from **week_6_set1_q5.csv** and display only the products whose stock is less than 10.

- **User Input:** No user input required.
- **Data Types:** Use string for product names, int for stock quantities.
- **File Reference:** Refer to **week_6_set1_q5.csv** for input.

Q6. Update Student Marks in a Dictionary Write a program to store student names and marks in a dictionary, update the marks of a given student, and display the updated dictionary.

- **User Input:** Student name (string) and new mark (int) entered by the user.
- **Data Types:** Use dict with string keys and int values.

Q7. Copy Lines Containing a Given Word into Another File Write a program to read a text file and copy only the lines containing a user-entered word into another text file.

- **User Input:** File name (string) and search word (string) entered by the user.
- **Data Types:** Use string for words and lines.
- **File Handling:** Use text file read mode (r) and write mode (w).

Q8. Write a Python program that asks the user to enter the names and ages of five people. Store the data in a list of lists and write the results into a CSV file named week_6_set1_q8.csv.

- **Data Types: string for names, int for ages.**

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